

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Architected a resilient **Puppeteer/Node.js automation engine** for MCA filings, reducing manual filing time by **85%** through network-idle handling, Vision API CAPTCHA solving, retries, and API-backed state checks.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized services with **Docker**, deployed to **Google Cloud Run**, and automated releases with **GitHub Actions**, improving deployment velocity by **70%**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built MRI segmentation tooling for **95 anatomical regions** using PyTorch/TinyGrad-compatible deep learning models and production-style preprocessing over large medical volumes.
- Optimized processing time by **87%** through asynchronous execution, parallel processing, and fewer repeated passes across high-dimensional image data.
- Extended 3D reconstructed surface visualization workflows, keeping model outputs usable for interactive inspection and downstream analysis.

Open Source

[Contributions](#)

- Contributed to Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, adapting patches to existing architecture, maintainer review, and project conventions.

Projects

Murdoc | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

[GitHub](#)

- Built an **AI gateway** for LLM, HTTP tool, and MCP traffic with route profiles, runtime policy checks, request scoring, and replayable decisions before downstream execution.
- Instrumented provider/tool flows with **OpenTelemetry** traces, Prometheus metrics, audit ledgers, and attack-lab validation so latency, failures, and unsafe actions could be investigated per request.

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, AWS, Prometheus*

[GitHub](#)

- Built observable Go services for a **content distribution platform** using browser automation, with RabbitMQ workers, DLQs, retries, Redis autosave, and PostgreSQL persistence.
- Added circuit-breaker logic and failure isolation for third-party publishing paths, keeping browser automation failures observable and recoverable instead of cascading through the system.

Distributed Inference Engine | *Python, PyTorch, FastAPI, gRPC, CUDA, Docker, Redis*

- Built a **distributed inference engine** to run models larger than a single consumer GPU by coordinating token generation across heterogeneous nodes with speculative decoding.
- Designed streaming APIs, worker heartbeats, retry-aware orchestration, and KV-cache-aware routing to balance latency, throughput, and worker availability across local and remote inference workers.

Technical Skills

Languages: Python, Go, TypeScript, C++

Frameworks: FastAPI, Echo, Node.js, Express, PyTorch, gRPC

Tools: Redis, PostgreSQL, Docker, Prometheus, OpenTelemetry, RabbitMQ